

SELENIUM LAB MANUAL

Common Setup (Do Once Before All Experiments)

1. Install Java JDK 11/17 (or higher) and set JAVA_HOME and Path.
2. Install Eclipse IDE for Java Developers.
3. Create a Maven Project in Eclipse (File > New > Maven Project).
4. In pom.xml, add Selenium dependency (selenium-java 4.x) and save.
5. Right-click project > Maven > Update Project (Force Update).
6. Create package: src/main/java > com.test.selenium.
7. Ensure Google Chrome or Microsoft Edge is installed and updated.

Note: Selenium Manager (Selenium 4.6+) usually downloads the correct driver automatically. So you normally do not need to download chromedriver.exe or set System.setProperty for drivers.

Experiment 1: Recording in Context Sensitive Mode and Analog Mode (Selenium IDE)

Tools/Requirements:

- Selenium IDE extension (Chrome/Edge)
- Any demo site for drag-and-drop (optional)

Procedure:

1. Open Chrome/Edge and install the Selenium IDE extension from the browser extensions store.
2. Click the Extensions icon and open Selenium IDE.
3. Click 'Create a new project' and enter a project name.
4. Click 'Add new test' and name the test case (e.g., TC_Record_Google).
5. Click the red 'Record' button.
6. Enter the base URL (e.g., <https://www.google.com>) and click 'Start Recording'.
7. Perform actions: click the search box, type a keyword, press Enter, click a result, then go back.
8. Stop recording and review the generated commands (open, click, type, assert).
9. Run the test using the 'Run current test' (play) button.

10. For analog-style actions: try recording a drag-and-drop or slider movement on a demo site and replay it.

11. Save the project in Selenium IDE.

Notes:

- Selenium IDE is mainly context-sensitive (element based). True analog recording is limited.
- For complex gestures use Selenium WebDriver code with the Actions class.

Output:

Running 'TC_Record_Google'

open | https://www.google.com |

type | name=q | selenium

assertTitle | Google |

Test completed successfully

Result: PASS

Experiment 2: GUI Checkpoint for Single Property (Title/Text/Attribute)

Tools/Requirements:

- Selenium WebDriver (Java)
- TestNG or JUnit (recommended)

Procedure:

1. Create a new Java class in package com.test.selenium (e.g., Exp2_SinglePropertyCheckpoint).
2. Launch the browser using Selenium WebDriver and open the target URL.
3. Choose one GUI property to verify (page title OR element text OR an attribute).
4. Capture the actual value (driver.getTitle() / element.getText() / element.getAttribute()).
5. Compare with expected value using assertion (TestNG/JUnit) or if-else logic.
6. Print PASS/FAIL in the console.
7. Close the browser.

Notes:

- Use WebDriverWait if the page/property loads slowly.

Sample Java Code:

```
// Exp 2: Single Property Checkpoint (Page Title)
package com.test.selenium;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class Exp2_SinglePropertyCheckpoint {
    public static void main(String[] args) throws InterruptedException {
        WebDriver driver = new ChromeDriver();
        driver.get("https://www.google.com");

        String expectedTitle = "Google";
        String actualTitle = driver.getTitle();

        System.out.println("Actual Title    : " + actualTitle);
        System.out.println("Expected Title : " + expectedTitle);

        if (actualTitle.equals(expectedTitle)) {
            System.out.println("PASS: Title is correct");
        } else {
            System.out.println("FAIL: Title is incorrect");
        }

        Thread.sleep(2000);
        driver.quit();
    }
}
```

Output

Actual Title : Google

Expected Title : Google

PASS: Title is correct

Experiment 3: GUI Checkpoint for Single Object/Window (Displayed/Enabled)

Tools/Requirements:

- Selenium WebDriver (Java)

Procedure:

1. Create a new Java class (e.g., Exp3_SingleObjectCheckpoint).
2. Launch browser and open the URL.

3. Identify the target element and choose a stable locator (id/name/css/xpath).
4. Use WebDriverWait to wait until the element is present/visible.
5. Verify one condition (isDisplayed / isEnabled / isSelected).
6. Print PASS/FAIL and close the browser.

Notes:

- Avoid Thread.sleep for synchronization; prefer explicit waits.

Sample Java Code:

```
// Exp 3: Single Object Checkpoint (Element displayed and enabled)
package com.test.selenium;

import org.openqa.selenium.*;
import org.openqa.selenium.chrome.ChromeDriver;

public class Exp3_SingleObjectCheckpoint {
    public static void main(String[] args) throws InterruptedException {
        WebDriver driver = new ChromeDriver();
        driver.get("https://www.google.com");

        WebElement searchBox = driver.findElement(By.name("q"));
        System.out.println("Displayed: " + searchBox.isDisplayed());
        System.out.println("Enabled : " + searchBox.isEnabled());

        System.out.println(searchBox.isDisplayed() && searchBox.isEnabled()
            ? "PASS: Search box is ready"
            : "FAIL: Search box is not ready");

        Thread.sleep(2000);
        driver.quit();
    }
}
```

Experiment 4(a): Bitmap Checkpoint for Object/Window (Screenshot Comparison)

Algorithm / Steps

1. Start the program.
2. Launch the Chrome browser.
3. Open the Google homepage.
4. Locate the Google Search Box using its name attribute (q).
5. Create a folder named **Screenshots** if it does not already exist.
6. Capture the screenshot of the search box object.

7. Save the screenshot as `current_searchbox.png`.
8. Check whether `baseline_searchbox.png` exists.
9. If the baseline image is not found:
 - Display a message asking the user to rename `current_searchbox.png` to `baseline_searchbox.png`.
10. If the baseline image exists:
 - Compare the baseline and current images pixel by pixel.
11. If both images are identical:
 - Display **BITMAP CHECKPOINT : TEST PASS**.
12. Otherwise:
 - Display **BITMAP CHECKPOINT : TEST FAIL**.
13. Close the browser.

```
package com.test.selenium;
import org.openqa.selenium.*;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.io.FileHandler;
import javax.imageio.ImageIO;
import java.awt.image.BufferedImage;
import java.io.File;
public class NewTest{
    public static void main(String[] args) throws Exception {
        WebDriver driver = new ChromeDriver();
        try {
            driver.get("https://www.google.com");
            // Locate object
            WebElement searchBox = driver.findElement(By.name("q"));
            // Create Screenshots folder
            File folder = new File("Screenshots");
            if (!folder.exists()) {
                folder.mkdir();
            }
            // Capture current screenshot
            File currentImage =
                searchBox.getScreenshotAs(OutputType.FILE);
            File currentFile =
                new File("Screenshots/current_searchbox.png");
            FileHandler.copy(currentImage, currentFile);
            System.out.println("Current screenshot saved:");
            System.out.println(currentFile.getAbsolutePath());
            // Baseline image
            File baselineFile =
                new File("Screenshots/baseline_searchbox.png");
            if (!baselineFile.exists()) {
                System.out.println("Baseline image not found.");
                System.out.println("Rename current_searchbox.png as baseline_searchbox.png");
            }
        }
        else {
```

```

        boolean result = compareImages(baselineFile, currentFile);
        if(result) {
            System.out.println("BITMAP CHECKPOINT : TEST PASS");
            System.out.println("Images are identical.")
        }

        else {
            System.out.println("BITMAP CHECKPOINT : TEST FAIL");
            System.out.println("Images are different.");
        }
    }

}

catch(Exception e) {
    System.out.println("Execution Failed");
    e.printStackTrace();
}

finally {
    driver.quit();
} }

// Method to compare image pixels
public static boolean compareImages(File img1, File img2)
    throws Exception {
    BufferedImage image1 = ImageIO.read(img1);
    BufferedImage image2 = ImageIO.read(img2);
    if(image1.getWidth()!=image2.getWidth() || image1.getHeight()!=image2.getHeight())
    {
        return false;
    }
    for(int x=0; x<image1.getWidth(); x++)
    {
        for(int y=0; y<image1.getHeight(); y++)
        {
            if(image1.getRGB(x,y) != image2.getRGB(x,y)
            return false;
        }
    }
    return true;
}

}

```

First Execution

Current screenshot saved:

C:\Users\YourName\eclipse-workspace\YourProject\Screenshots\current_searchbox.png

Baseline image not found.

Rename current_searchbox.png as baseline_searchbox.png

Second Execution

Current screenshot saved:

C:\Users\YourName\eclipse-workspace\YourProject\Screenshots\current_searchbox.png

BITMAP CHECKPOINT : TEST PASS

Images are identical.

Experiment 4(b): Bitmap Checkpoint for Screen Area (Crop + Compare)

Algorithm / Steps

1. Start the program.
2. Launch the Chrome browser.
3. Maximize the browser window.
4. Open the Google homepage.
5. Wait for 3 seconds to allow the page to load completely.
6. Capture a screenshot of the entire browser window.
7. Convert the screenshot into a `BufferedImage`.
8. Create a folder named `C:\SeleniumScreenshots` if it does not already exist.
9. Crop the top-left screen area of size **800 × 600 pixels**.
10. Save the cropped image as `current_google_area.png`.
11. Check whether `baseline_google_area.png` exists.
12. If the baseline image does not exist:
 - Display a message asking the user to rename `current_google_area.png` as `baseline_google_area.png`.
13. If the baseline image exists:
 - Compare the baseline image and the current image pixel by pixel.
 - Calculate the similarity percentage.
14. If the similarity is **95% or more**, display **TEST PASS**.
15. Otherwise, display **TEST FAIL**.
16. Close the browser.

17. End the program.

```
import javax.imageio.ImageIO;
import org.openqa.selenium.*;
import org.openqa.selenium.chrome.ChromeDriver;
public class NewTest {
public static void main(String[] args) throws Exception {
WebDriver driver = new ChromeDriver();
try {
    driver.manage().window().maximize();
    driver.get("https://www.google.com");
    Thread.sleep(3000);
    // Capture screenshot
        File screenshot=((TakesScreenshot) driver).getScreenshotAs(OutputType.FILE);
    BufferedImage fullImage = ImageIO.read(screenshot);
    // Local folder
    File folder = new File("C:\\SeleniumScreenshots");
    if (!folder.exists())
    {
        folder.mkdirs();
    }
    // Capture screen area
    int width = Math.min(800, fullImage.getWidth());
    int height = Math.min(600, fullImage.getHeight());
    BufferedImage currentArea = fullImage.getSubimage(0,0,width,height);
    File currentFile = new File("C:\\SeleniumScreenshots\\current_google_area.png");
    ImageIO.write(currentArea, "png", currentFile);
    System.out.println("Current screenshot saved:");
    System.out.println(currentFile.getAbsolutePath());
    File baselineFile = new File("C:\\SeleniumScreenshots\\baseline_google_area.png");
    if(!baselineFile.exists()) {
    System.out.println("Baseline image not found.");
    System.out.println("Rename current_google_area.png to baseline_google_area.png");
    }
    else {
        BufferedImage baseline = ImageIO.read(baselineFile);
        double similarity =compareImages(baseline, currentArea);
        System.out.println("Image Similarity : " + similarity + "%");
        if(similarity >= 95) {
            System.out.println("BITMAP CHECKPOINT : TEST PASS");
        }
    }
    else {
```



```

System.out.println("BITMAP CHECKPOINT : TEST FAIL");
}
}

    }
    finally {
        driver.quit();
    }
}

public static double compareImages( BufferedImage img1,BufferedImage img2)
{
    if(img1.getWidth()!=img2.getWidth() || img1.getHeight()!=img2.getHeight()) {
        return 0;
    }
    long totalPixels = img1.getWidth()* img1.getHeight();
    long samePixels = 0;
    for(int x=0; x<img1.getWidth(); x++)
    {
        for(int y=0; y<img1.getHeight(); y++)
        {
            if(img1.getRGB(x,y) == img2.getRGB(x,y))
            {
                samePixels++;
            }
        }
    }
    return
        ((double)samePixels / totalPixels) * 100;

}

}

```

Output

First Execution

Current screenshot saved:

C:\SeleniumScreenshots\current_google_area.png

Baseline image not found.

Rename current_google_area.png to baseline_google_area.png

Second Execution

Current screenshot saved:

C:\SeleniumScreenshots\current_google_area.png

Image Similarity : 99.72%

BITMAP CHECKPOINT : TEST PASS

Experiment 5: Database Checkpoint – Default Check (Record Exists)

// Not For Students (for your reference)

To install H2 DATABASE Dependency for pom.xml

```
<dependency>
  <groupId>com.h2database</groupId>
  <artifactId>h2</artifactId>
  <version>2.4.240</version>
</dependency>
```

```
package com.test.selenium;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
public class NewTest {
public static void main(String[] args) {
try {
  // Connect to H2 Database
  Connection con =
DriverManager.getConnection("jdbc:h2:./StudentDB","sa","");
  Statement stmt = con.createStatement();
  // Create table if it does not exist
  stmt.execute("CREATE TABLE IF NOT EXISTS STUDENT (" + "ID INT PRIMARY KEY,
"+ "NAME VARCHAR(50)," + "EMAIL VARCHAR(100))");
  // Remove previous data (optional)
  stmt.execute("DELETE FROM STUDENT");
  // Insert one record
  stmt.executeUpdate("INSERT INTO STUDENT VALUES " +
"(1,'Ajay','ajay@gmail.com')");
  // Database Checkpoint
  ResultSet rs = stmt.executeQuery("SELECT * FROM STUDENT WHERE ID = 1");
  if (rs.next()) {
```

```

        System.out.println("Record Exists");
        System.out.println("ID    : " + rs.getInt("ID"));
        System.out.println("Name  : " + rs.getString("NAME"));
        System.out.println("Email : " + rs.getString("EMAIL"));
        System.out.println("TEST PASSED");
    } else {
        System.out.println("Record Not Found");
        System.out.println("TEST FAILED");
    }
}
// Close connection
    con.close();
} catch (Exception e) {
    e.printStackTrace();
}
}
}

```

Output

```

Record Exists
ID   : 1
Name : Ajay
Email : ajay@gmail.com
TEST PASSED

```

Process finished with exit code 0

=====

Experiment 6: Database Checkpoint – Custom Check (Validate Specific Value)

```

package com.test.selenium;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
public class NewTest{
    public static void main(String[] args) {

        try {

            // Connect to H2 Database

```

```

Connection con = DriverManager.getConnection(
    "jdbc:h2:./StudentDB", "sa", "");

Statement stmt = con.createStatement();

// Create table if it does not exist
stmt.execute("CREATE TABLE IF NOT EXISTS STUDENT ( "
    + "ID INT PRIMARY KEY, "
    + "NAME VARCHAR(50), "
    + "EMAIL VARCHAR(100))");

// Delete previous records
stmt.execute("DELETE FROM STUDENT");

// Insert sample record
stmt.executeUpdate(
    "INSERT INTO STUDENT VALUES "
    + "(1, 'Ajay', 'ajay@gmail.com')");

// Retrieve the record
ResultSet rs = stmt.executeQuery(
    "SELECT NAME FROM STUDENT WHERE ID = 1");

if (rs.next()) {

    String actualName = rs.getString("NAME");
    String expectedName = "Ajay";

    // Custom Check
    if (actualName.equals(expectedName)) {

        System.out.println("Expected Name : " + expectedName);
        System.out.println("Actual Name : " + actualName);
        System.out.println("CUSTOM CHECK PASSED");

    } else {

        System.out.println("Expected Name : " + expectedName);
        System.out.println("Actual Name : " + actualName);
        System.out.println("CUSTOM CHECK FAILED");
    }

} else {

```

```

        System.out.println("Record Not Found");
    }

    con.close();

    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

Output

Expected Name : Ajay
 Actual Name : Ajay
 CUSTOM CHECK PASSED

Process finished with exit code 0

=====

Experiment 7(a): Data Driven Test – Dynamic Test Data Submission

Create Html file

```

<!DOCTYPE html>
<html>
<head>
    <title>Student Registration</title>
</head>
<body>
<h2>Student Registration Form</h2>
<label>Name:</label>
<input type="text" id="name"><br><br>
<label>Email:</label>
<input type="text" id="email"><br><br>
<button onclick="submitForm()">Submit</button>
<p id="result"></p>
<script>
function submitForm(){
var name=document.getElementById("name").value;
var email=document.getElementById("email").value;
document.getElementById("result").innerHTML=
"Registration Successful";

```

```
}  
</script>  
</body>  
</html>
```

Save the Html file in the location which is in BOLD in program

```
package com.test.selenium;  
import org.openqa.selenium.*;  
import org.openqa.selenium.chrome.ChromeDriver;  
public class DynamicDataSubmission {  
    public static void main(String[] args)  
        throws Exception {  
  
        WebDriver driver = new ChromeDriver();  
  
        // Open local HTML page  
        driver.get("file:///C:/SeleniumFiles/StudentForm.html");  
  
        // Dynamic Test Data  
        String[][] data = {  
  
            {"Ajay", "ajay@gmail.com"},  
            {"Rahul", "rahul@gmail.com"},  
            {"Anita", "anita@gmail.com"}  
  
        };  
  
        for(int i=0;i<data.length;i++)  
        {  
  
            WebElement name = driver.findElement(By.id("name"));  
  
            WebElement email = driver.findElement(By.id("email"));  
            name.clear();  
            email.clear();  
            name.sendKeys(data[i][0]);  
            email.sendKeys(data[i][1]);  
  
            driver.findElement(By.tagName("button")).click();  
  
            String result = driver.findElement(  
                By.id("result")).getText();
```

```

        System.out.println("-----");
        System.out.println("Test Case : "+(i+1));
        System.out.println("Name : "+data[i][0]);
        System.out.println("Email : "+data[i][1]);
        System.out.println(result);
        if(result.equals("Registration Successful"))
            System.out.println("TEST PASS");
        else
            System.out.println("TEST FAIL");

        Thread.sleep(1000);
    }
    driver.quit();
}
}

```

Output

```

Test Case : 1
Name : Ajay
Email : ajay@gmail.com
Registration Successful
TEST PASS
-----
Test Case : 2
Name : Rahul
Email : rahul@gmail.com
Registration Successful
TEST PASS
-----
Test Case : 3
Name : Anita
Email : anita@gmail.com
Registration Successful
TEST PASS

```

=====

Experiment 7(b):Data Driven Test – Through Flat Files (CSV/TXT)

Step 1: Create a HTML File(StudentForm.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration Form</title>
</head>
<body>
<h2>Student Registration Form</h2>
<label>Name :</label>
<input type="text" id="name">
<br><br>
<label>Email :</label>
<input type="text" id="email">
<br><br>
<button onclick="submitForm()">Submit</button>
<p id="result"></p>
<script>
function submitForm()
{
  var name=document.getElementById("name").value;
  var email=document.getElementById("email").value;

  if(name!="" && email!="")
  {
    document.getElementById("result").innerHTML=
      "Registration Successful";
  }
  else
  {
    document.getElementById("result").innerHTML=
      "Registration Failed";
  }
}
</script>
</body>
</html>
```


Step 2: Create CSV file with data(StudentData.csv)

Ajay,ajay@gmail.com

Rahul,rahul@gmail.com

Anita,anita@gmail.com

Step 3 Selenium Code

```
package com.test.selenium;
import java.io.BufferedReader;
import java.io.FileReader;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
public class DataDrivenCSV {
    public static void main(String[] args) throws Exception {
        // Launch Chrome
        WebDriver driver = new ChromeDriver();

        // Maximize browser
        driver.manage().window().maximize();

        // Open HTML page
        driver.get("file:///C:/SeleniumFiles/StudentForm.html");

        // Read CSV file
        BufferedReader br =
            new BufferedReader(
                new FileReader(
                    "C:\\SeleniumFiles\\StudentData.csv"));

        String line;

        int testCase = 1;

        while ((line = br.readLine()) != null) {

            // Split CSV data
            String[] data = line.split(",");

            // Locate elements
```

```
WebElement name =
    driver.findElement(By.id("name"));

WebElement email =
    driver.findElement(By.id("email"));

// Clear old values
name.clear();
email.clear();

// Enter data
name.sendKeys(data[0]);
email.sendKeys(data[1]);

// Click Submit
driver.findElement(By.tagName("button")).click();

// Read Result
String result =
    driver.findElement(By.id("result")).getText();

// Display Output
System.out.println("-----");

System.out.println("Test Case : " + testCase);

System.out.println("Name : " + data[0]);

System.out.println("Email : " + data[1]);

System.out.println("Expected Result : Registration Successful");

System.out.println("Actual Result : " + result);

if(result.equals("Registration Successful"))
{
    System.out.println("TEST PASS");
}
else
{
    System.out.println("TEST FAIL");
}

testCase++;
```

```
        Thread.sleep(1500);

    }

    br.close();

    driver.quit();

}

}
```

Output

```
Test Case : 1
Name : Ajay
Email : ajay@gmail.com
Expected Result : Registration Successful
Actual Result : Registration Successful
TEST PASS
-----
Test Case : 2
Name : Rahul
Email : rahul@gmail.com
Expected Result : Registration Successful
Actual Result : Registration Successful
TEST PASS
-----
Test Case : 3
Name : Anita
Email : anita@gmail.com
Expected Result : Registration Successful
Actual Result : Registration Successful
TEST PASS
```

=====

Experiment 7(c): Data Driven Test – Through Front Grids (Web Tables)

Create Html file(web_table)

```
<!DOCTYPE html>
<html>
<head>
  <title>Front Grid Example</title>
</head>
<body>
<h2>Student Data (Front Grid)</h2>
<table border="1" id="studentTable">
<tr>
<th>Name</th>
<th>Email</th>
</tr>

<tr>
<td>Ajay</td>
<td>ajay@gmail.com</td>
</tr>

<tr>
<td>Rahul</td>
<td>rahul@gmail.com</td>
</tr>

<tr>
<td>Anita</td>
<td>anita@gmail.com</td>
</tr>

</table>
<br><br>
<h2>Registration Form</h2>
Name :
<input type="text" id="name">
<br><br>

Email :
<input type="text" id="email">
<br><br>
<button onclick="submitForm()">
Submit
</button>
<p id="result"></p>
<script>
```

```

function submitForm(){
document.getElementById("result").innerHTML=
"Registration Successful";
}
</script>
</body>
</html>

```

Selenium Code

```

package com.test.selenium;

import java.util.List;

import org.openqa.selenium.*;
import org.openqa.selenium.chrome.ChromeDriver;

public class DataDrivenFrontGrid {

    public static void main(String[] args)
        throws Exception {

        WebDriver driver = new ChromeDriver();

        driver.manage().window().maximize();

        driver.get("file:///C:/SeleniumFiles/WebTableForm.html");

        // Read all rows except header
        List<WebElement> rows =
            driver.findElements(
                By.xpath("//table[@id='studentTable']/tbody/tr"));

        for(int i=2;i<=rows.size();i++) {

            String name =
                driver.findElement(
                    By.xpath("//table[@id='studentTable']/tbody/tr["+i+"]/td[1]"))
                    .getText();

            String email =
                driver.findElement(
                    By.xpath("//table[@id='studentTable']/tbody/tr["+i+"]/td[2]"))

```

```

        .getText();

WebElement txtName =
    driver.findElement(By.id("name"));

WebElement txtEmail =
    driver.findElement(By.id("email"));

txtName.clear();
txtEmail.clear();

txtName.sendKeys(name);
txtEmail.sendKeys(email);

driver.findElement(
    By.tagName("button")).click();

String result =
    driver.findElement(
        By.id("result")).getText();

System.out.println("-----");

System.out.println("Name : " + name);

System.out.println("Email : " + email);

System.out.println(result);

if(result.equals("Registration Successful"))
    System.out.println("TEST PASS");
else
    System.out.println("TEST FAIL");

Thread.sleep(1500);

}

driver.quit();

}

}

```

OutPut

Name : Ajay
Email : ajay@gmail.com
Registration Successful
TEST PASS

Name : Rahul
Email : rahul@gmail.com
Registration Successful
TEST PASS

Name : Anita
Email : anita@gmail.com
Registration Successful
TEST PASS

=====

Experiment 7(d): Data Driven Test – Through Excel

Step 1 Create Html file (Studentform.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration</title>
</head>
<body>

<h2>Student Registration Form</h2>

<label>Name :</label>
<input type="text" id="name"><br><br>

<label>Email :</label>
<input type="text" id="email"><br><br>

<button onclick="submitForm()">Submit</button>

<p id="result"></p>

<script>
```

```

function submitForm(){

var name=document.getElementById("name").value;
var email=document.getElementById("email").value;

if(name!="" && email!="")
{
document.getElementById("result").innerHTML=
"Registration Successful";
}
else
{
document.getElementById("result").innerHTML=
"Registration Failed";
}

}
</script>

</body>
</html>

```

Step 2 Create Excel file (StudentData.xlsx) with Data

Name	Email
Ajay	ajay@gmail.com
Rahul	rahul@gmail.com
Anita	anita@gmail.com
Priya	priya@gmail.com

Step 3: if the Apache POI JAR either by adding Maven Dependency or as external jars

For Maven

```

<dependency>
  <groupId>org.apache.poi</groupId>
  <artifactId>poi-ooxml</artifactId>
  <version>5.2.5</version>
</dependency>

```


Step 4 :

```
package com.test.selenium;

import java.io.File;
import java.io.FileInputStream;

import org.apache.poi.ss.usermodel.*;
import org.apache.poi.xssf.usermodel.XSSFWorkbook;

import org.openqa.selenium.*;
import org.openqa.selenium.chrome.ChromeDriver;

public class DataDrivenExcel {

    public static void main(String[] args) throws Exception {

        // Open Excel File
        File file =
            new File("C:\\SeleniumFiles\\StudentData.xlsx");

        FileInputStream fis =
            new FileInputStream(file);

        Workbook workbook =
            new XSSFWorkbook(fis);

        Sheet sheet =
            workbook.getSheetAt(0);

        WebDriver driver =
            new ChromeDriver();

        driver.manage().window().maximize();

        driver.get("file:///C:/SeleniumFiles/StudentForm.html");

        // Start from row 1 because row 0 contains headers
        for(int i=1; i<=sheet.getLastRowNum(); i++) {

            Row row = sheet.getRow(i);

            String name =
```

```

        row.getCell(0).getStringCellValue();

String email =
    row.getCell(1).getStringCellValue();

WebElement txtName =
    driver.findElement(By.id("name"));

WebElement txtEmail =
    driver.findElement(By.id("email"));

txtName.clear();
txtEmail.clear();

txtName.sendKeys(name);
txtEmail.sendKeys(email);

driver.findElement(
    By.tagName("button")).click();

String result =
    driver.findElement(
        By.id("result")).getText();

System.out.println("-----");
System.out.println("Test Case : " + i);
System.out.println("Name : " + name);
System.out.println("Email : " + email);
System.out.println("Expected Result : Registration Successful");
System.out.println("Actual Result : " + result);

if(result.equals("Registration Successful"))
    System.out.println("TEST PASS");
else
    System.out.println("TEST FAIL");

Thread.sleep(1000);

}

workbook.close();
fis.close();

driver.quit();

```

```
}  
  
}
```

Output :

```
Test Case : 1  
Name : Ajay  
Email : ajay@gmail.com  
Expected Result : Registration Successful  
Actual Result : Registration Successful  
TEST PASS
```

```
-----  
Test Case : 2  
Name : Rahul  
Email : rahul@gmail.com  
Expected Result : Registration Successful  
Actual Result : Registration Successful  
TEST PASS
```

```
-----  
Test Case : 3  
Name : Anita  
Email : anita@gmail.com  
Expected Result : Registration Successful  
Actual Result : Registration Successful  
TEST PASS
```

```
-----  
Test Case : 4  
Name : Priya  
Email : priya@gmail.com  
Expected Result : Registration Successful  
Actual Result : Registration Successful  
TEST PASS
```

=====

For Experiments 7(e),7(f) ,8 Testng is required to install

For Testng jars update this Dependency in pom.xml

```
<dependency>  
  <groupId>org.testng</groupId>  
  <artifactId>testng</artifactId>  
  <version>7.11.0</version>
```

</dependency>

For Testng launcher if not installed in eclipse

In help-→Go to marketplace-→ search for Testng and install it

Experiment 7(e): Batch Testing – Without Parameter Passing (Suite Run)

Step 1 Create 3 Test Classes

Class 1

```
package com.test.selenium;
import org.testng.annotations.Test;
public class TestBing {
    @Test
    public void gmailTest()
{
        System.out.println("Bing test executed");
    }
}
```

Class 2

```
package com.test.selenium;
import org.testng.annotations.Test;

public class TestGoogle {
    @Test
    public void googleTest() {
        System.out.println("Google Test Executed");
    }
}
```

Class 3

```
package com.test.selenium;
import org.testng.annotations.Test;
```

```

public class TestYahoo {

    @Test
    public void youtubeTest() {

        System.out.println("Test yahoo executed");

    }

}

```

Step 2 Create testing.xml file

Right click on your java project --> new----> file---->give file name (testing.xml) -->finish

Paste this in testing.xml

```

<!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">

<suite name="BatchSuite">

    <test name="WithoutParameterPassing">

        <classes>

            <class name="com.test.selenium.TestBing"/>

            <class name="com.test.selenium.TestGoogle"/>

            <class name="com.test.selenium.TestYahoo"/>

        </classes>

    </test>

</suite>

```

Now after saving this xml -> right click on testing.xml in project explorer-->Run as Testng suite

Output

```

Bing test executed
Google Test Executed

```

Test yahoo executed

```
=====
BatchSuite
Total tests run: 3, Passes: 3, Failures: 0, Skips: 0
=====
```

Experiment 7(f): Batch Testing – With Parameter Passing (Suite Parameters)

Step 1 Create java class TestParameter.java

```
package com.test.selenium;

import org.testng.annotations.Parameters;
import org.testng.annotations.Test;

public class TestParameter {

    @Parameters({"username","password"})
    @Test
    public void login(String user, String pass) {

        System.out.println("Username : " + user);
        System.out.println("Password : " + pass);

        if(user.equals("admin") && pass.equals("admin123")) {

            System.out.println("TEST PASS");

        } else {

            System.out.println("TEST FAIL");

        }

    }

}
```

Step 2 Create new xml file testng1.xml

Right click on your java project --> new----> file---->give file name (testing1.xml) -->finish

Paste this in testing1.xml

```
<!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">
```

```
<suite name="ParameterSuite">
```

```
  <test name="LoginTest">
```

```
    <parameter name="username" value="admin"/>
```

```
    <parameter name="password" value="admin123"/>
```

```
    <classes>
```

```
      <class name="com.test.selenium.TestParameter"/>
```

```
    </classes>
```

```
  </test>
```

```
</suite>
```

Now after saving this xml -> right click on testing1.xml in project explorer-->Run as Testng suite

Output:

```
Username : admin  
Password : admin123  
TEST PASS
```

```
=====
```

```
ParameterSuite
```

```
Total tests run: 1, Passes: 1, Failures: 0, Skips: 0
```

```
=====
```

Experiment 8: Data Driven Batch (Batch + Multiple Data Sets)

Step 1 Create java class LoginTest.java

```
package com.test.selenium;
```

```

import org.testng.annotations.DataProvider;
import org.testng.annotations.Test;

public class LoginTest {

    @DataProvider(name="LoginData")

    public Object[][] getData() {

        return new Object[][] {

            {"admin","admin123"},

            {"student","student123"},

            {"guest","guest123"}

        };

    }

    @Test(dataProvider="LoginData")

    public void login(String user,String pass) {

        System.out.println("Username : " + user);

        System.out.println("Password : " + pass);

        System.out.println("Login Successful");

        System.out.println("-----");

    }

}

```

Output:

```

Username : admin
Password : admin123
Login Successful

```



```

-----
Username : student
Password : student123
Login Successful
-----
Username : guest
Password : guest123
Login Successful
-----
PASSED: com.test.selenium.LoginTest.login("admin", "admin123")
PASSED: com.test.selenium.LoginTest.login("student", "student123")
PASSED: com.test.selenium.LoginTest.login("guest", "guest123")

=====
    Default test
    Tests run: 1, Failures: 0, Skips: 0
=====

=====
Default suite
Total tests run: 3, Passes: 3, Failures: 0, Skips: 0
=====

=====

```

Experiment 9: Silent Mode Test Execution (Headless)

```

package com.test.selenium;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;

public class SilentModeExecution {

    public static void main(String[] args) {

        // Create Chrome options
        ChromeOptions options = new ChromeOptions();

        // Enable Headless Mode
        options.addArguments("--headless=new");
    }
}

```

```
// Optional: Set browser window size
options.addArguments("--window-size=1920,1080");

// Launch Chrome in Headless Mode
WebDriver driver = new ChromeDriver(options);

// Open Google
driver.get("https://www.google.com");

// Find search box
WebElement searchBox = driver.findElement(By.name("q"));

// Enter text
searchBox.sendKeys("Selenium WebDriver");

// Get page title
String title = driver.getTitle();

System.out.println("Page Title : " + title);

// Checkpoint
if (title.contains("Google")) {

    System.out.println("TEST PASS");

} else {

    System.out.println("TEST FAIL");

}

// Close browser
driver.quit();
}
}
```

OutPut

Page Title : Google
TEST PASS